GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Instrument Design Basis**
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Instrument Design Basis

R03	05.08.22	Re-issued for Review	P. Roy	P. Olukah	A. Duru	
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Revision History

Rev No.	Date	Description of Revision
R01	27.06.22	Issued for Review
R02	07.07.22	Re-issued for Review
R03	05.08.22	Re-issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-023

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

Table of Contents

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
IEC	International Electro technical Commission
ISA	Instrument Society of America
ITU	International telecommunication Union
SCADA	Supervisory Control and Data Acquisition System
barg	Bar Gauge (Pressure)
bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
Bcf	Billion Cubic Feet
Bopd	Barrels of Oil Per Day
Bpd or B/D	Barrel Per Day
BS&W	Base Sediment and Water
Bwpd	Barrels of Water Per Day
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	Kilometre
m	Metre
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousands Barrels Per Day
Mbwpd	Thousand Barrels of Water Per Day
MJ	Megajoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Parts Per Million

psig	Pounds Per Square Inch Gauge (Pressure)

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

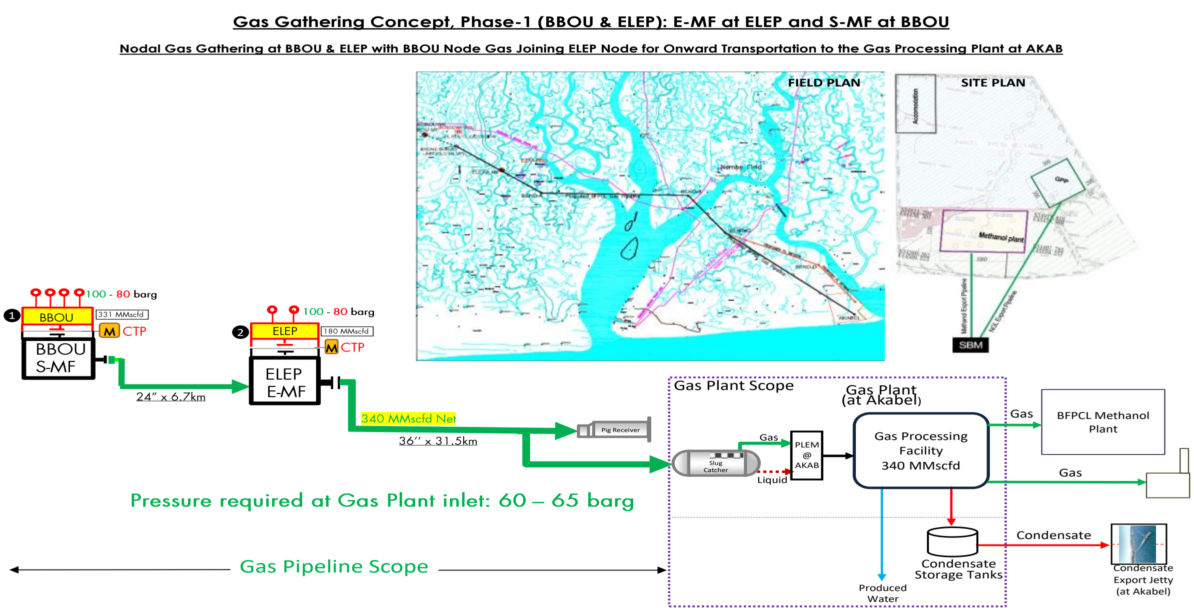

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document defines the basic requirements for the engineering and design of the instrumentation, control and safety. The FEED shall fully comply with all specified relevant contractual requirements. This document defines the basic requirements for the engineering and design of the instrumentation for BBOU, ELEP and AKAB pipeline. The purpose of this specification is to ensure a standard approach is maintained to the engineering and design of instrumentation and equipment and that it complies with all relevant national and international standards.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General

3.1. Instruments and equipment shall be consistent with the environmental impact statement, with particular emphasis to the following:

Noise within 20 meters of the battery limit must not exceed 45dBA. Fugitive process emissions must be minimized e.g.) from control valve glands and analyser vents.

3.2. Field instrumentation shall have a minimum ingress protection of IP65 in accordance with IEC 60529. Ingress protection for Level Gauge Indicator shall be IP-67. All instruments shall be intrinsically safe or explosion proof and shall be certified to specific hazardous area class by statutory body such as ATEX, PTB.

3.3. Materials of Construction

Materials of construction for instrument equipment shall be consistent with corrosion resistance and the pressure and temperature conditions of the process fluids involved. Generally the parts in contact with the process fluid shall be of a material equal to or better than that used for the related vessel or line. Where instrument design prohibits the use of these materials, or where a suitable material cannot be reasonably procured, the instruments shall be isolated from the process by means of a chemical seal which shall be identified on P&ID.

Wetted parts including sensing elements and also movement parts shall be of AISI 316 stainless steel. The material employed in special services shall be selected according to application from SS316L, Titanium, Monel, Hastelloy and etc. The materials of wetted part in sour services shall generally be selected and treated in accordance with the recommendation of NACE MR 0103

The following materials shall not be used in contact with the product:

Aluminium for wetted parts, Zinc,Asbestos,Brass, Bronze, Copper. Heavy metals including Mercury, Antimony and Cadmium. Plastics containing free Phenol, formaldehyde or plasticizers.

3.4. Pressure & temperature rating

Pressure and temperature rating of instruments and components shall conform to the design rating of the process system to which they are connected. Instruments which can be exposed to vacuum shall have under-range protection to full vacuum.

3.5. Hazardous area requirement

3.5.1. Classification of Hazardous Areas shall be Zone1, IIA/IIB, T4.

All instrumentation located or partially located in the hazardous area shall be suitable for classified hazardous area. All instruments and Junction Boxes shall be certified as Explosion Proof (Exd). Power supply shall be switched off before opening the Instrument or Junction Box. ATEX/PTB approved certification shall be supplied for all electrical / electronic equipment in the hazardous area.

3.6. Accessibility: All instruments shall be accessible from grade or platform for measurement .In case of gas service where instruments are mounted above the taps, the installation of platform shall be considered, if accessibility is not available.

3.7. Analogue signals

3.7.1. Transmitters shall be two-wire 4-20 mA dc powered, microprocessor based, SMART (HART protocol) with local LCD indication.

3.7.2. In case of HART transmitters, the minimum requirement shall be as follows.

3.7.2.1. Communication Protocol shall be HART(Latest Version) and shall be suitable for HART Maintenance System.

3.7.2.2. Accuracy of transmitters shall be +0.1% or range or better for a rangeability of 1:15. The accuracy includes the combined effect of linearity, hysteresis and repeatability.

3.7.2.3. Hand held configuration shall be provided for remote calibration, remote configuration and diagnostics. It shall be possible to connect the configurator in the main control room or junction box located in field. One (1) No. Hand held communicator shall be provided for BBOU,ELEP and AKAB each platform.

3.7.3. Transmitters shall be supplied with integral output meter (LCD type).

3.7.4. Transmitter accuracy, repeatability and rangeability shall be suitable for their intended use.

3.8. Digital signals

3.8.1. Position switches for ESD/BDV valves shall be two-wire proximity type.

3.8.2. Outputs to solenoid valves shall be 24Vdc.

3.9. Restriction Orifice Plates

3.9.1. Restriction Orifice plates shall be used in the flare line/vent line. Thickness of the plate shall be suitable for pressure condition and sonic velocity consideration of the flare line. Thickness of the Restriction orifice plate shall be calculated based on the R W Miller equation.

4. Temperature Instruments

4.1. Temperature elements

4.1.1. Where the measuring temperature ≤ 400°C," RTD (Platinum, 100 ohm at 0 °C) complying with IEC-60751 Class 'A' shall be used. '4 wire' RTD for check metering unit shall be acceptable, if considered by OEM.

4.2. Transmitters

4.2.1. Microprocessor based, SMART (4-20 mA with HART) with local LCD indication shall be used.

4.2.2. Transmitter shall be integrally mounted with RTD. Accuracy of the Temperature Transmitter shall be +/-0.08 degC.

4.2.3. In the event of temperature element failure, transmitters shall be arranged to drive towards the alarm condition or safe control condition where one exists, or otherwise upscale.

4.3. Protecting Thermowells

4.3.1. Thermowell shall be machined from bar stock material. Other materials shall be furnished as required in the piping specification. Temperature elements shall be supplied complete with thermowell, extension, head and integrally mounted transmitter as a complete assembly. All thermowells shall be SS316L and meet the Karman Wake Frequency calculation as per PTC 19.3. If it fails to achieve the wake frequency calculation, collar or vortex shedding design shall be used to meet the velocity criteria. Thermowell ts design and geometry shall meet the requirements of Part II, Section 3.9 of DEP 31.38.01.24-Gen. At all process conditions, the ratio of the Strauhal frequency and the thermowell natural frequency shall be less than 0.8, as stated in ASME PTC 19.3.

4.3.2. Thermowells shall be flanged according to equipment and pipe specifications. The flange size shall be 40 mm (1.1/2 inch) with rating and facing according to the line classification.

4.3.3. According to API RP 551, the immersion length of thermowells in the pipes (the length from well tip to pipe wall) shall be maximum 5" or 125mm. The immersion length in vessels liquid shall be 400 mm. In vessels containing gas, the immersion length shall be according to the table below:

Vessel diameter (D)	Thermowell immersion length(mm)
D upto 800 mm	200
800<D≤1200 mm	300

The insertion length depends on the length of the nozzle or pipe fitting, taking any possible refractory lining into account.

If the thermowell design fails to pass the wake frequency with the give gas velocity even after increasing the tip diameter or other mechanical parameters, vortex shedding type helical thermowell shall be used. Velocity collar shall be avoided.

4.4. Local temperature indicators

4.4.1. Local temperature indicators shall be rigid stem bimetallic dial thermometer type supplied complete with protective thermowells with specification mentioned above.

4.4.2. Dial size shall be 150 mm nominal and instruments shall be heavy duty `every angle' head type. Temperature scales shall be direct reading and comply with manufacturer's standard ranges. The Case shall be SS304 and Stem shall be SS316.

4.4.3. Scales shall be selected to cover the range requirements for both normal and design conditions of plant operation Mercury filled systems must not be used.

5. Pressure Instruments

5.1. Pressure transmitters

5.1.1. Pressure transmitters shall be referenced to atmospheric pressure.

5.1.2. Transmitters shall be microprocessor based smart (4-20mA HART) type, with display unit and independent adjustment of zero and span at the instrument and from a remote location. Accuracy shall be +/-0.075% of the calibrated span.

5.1.3. Materials of construction shall be suitable for the process media and based upon the piping specification. In general pressure transmitters should have Aluminium Enclosure with low copper and stainless-steel wetted pats.

5.1.4. Scales shall be selected to cover the range requirements for both start-up, normal and design conditions of plant operation.

5.2. Pressure gauges

5.2.1. Pressure gauge accuracy shall be + 1.0% of full range.

5.2.2. Dial size shall be 150 mm and cases shall be stainless steel screw on or bayonet bezel type. Blowout disc protection shall be provided and gauges shall be orientated such that they vent safely.

5.2.3. Gauge windows shall be constructed from safety pattern / toughened glass.

5.2.4. Gauges shall be normally Bourdon tube type. Bellows or diaphragm-actuated gauges shall be used for lower pressure ranges. Chemical seals with body and diaphragm in stainless steel shall be used if the medium tends to crystallise, is highly viscous, or where suspended matter exists. The process connection of flanged pressure gauge should be 2” flange.

5.2.5. Bourdon tube material shall be type 316 stainless steel, as a minimum, with stainless steel movement, except where process conditions require more stringent materials. Martensitic stainless steel must be used for gases containing hydrogen.

5.2.6. Over-range protection shall be 1.3 times the maximum scale range. Where a gauge is subject a greater pressure, a gauge saver shall be used. All pressure gauges installed on the Pig Launcher and Pig Receiver shall have Gauge Saver.

5.2.7. A gauge valve shall be provided for isolation and calibration purposes. A primary isolation valve shall be provided where shown in the P&ID.

5.2.8. Gauges shall be connected to the piping by means of Mono-flanges.

5.2.9. Gauge connections shall be 1/2" NPTM.

5.3. Instrument Manifolds

5.3.1. 2-valve manifolds shall be used for secondary isolation, venting and calibration of pressure instruments.

5.3.2. 5-valve manifolds shall be used for secondary isolation, venting and calibration of differential pressure instruments

5.3.3. Alternative to 2-valve and 5-valve manifold, mono-flange with double block and bleed (DBB) can be used. Mono-flange shall have ½" NPTF connection for pressure gauge and pressure transmitter connection. Mono-flange assembly shall be provided with closed-coupled hook-up. For unmanned platforms, mono-flange assembly shall help in reducing/elimination impulse tubing that could be source of leakage.

6. Flow Meters (Wet Gas Metering System)

6.1. Venturi Meter shall be used as 'CHECK' meter. Venturi based wet gas meters shall be designed based on ISO 5167, ISO 11583, ASME MFC 19G, ISO TR 12748 and other related international standards.

6.2. Low powered Field mounted Flow Computer (Hazardous Area) along with all Input/Output modules that shall interfaces with the PT,DPT and TT from the venturi measurement. All necessary software shall be supplied and configured. Interface with SCADA system through serial links.

6.3. The scope of venture based wet gas metering system for BBOU and ELEP shall consist of following

- Design of the wet gas metering system based on the process data provided in this specification.
- Supply of integrated assembly of venture meters along with Gauge Pressure Transmitters (PT), Differential Pressure Transmitters (DPT), Temperature Transmitter with RTD and TW, First isolation valve for pressure and differential transmitters, Double Block and Bleed Valve Pressure and differential pressure transmitters, impulse piping and fitting complete with installation of all transmitters.

6.4. All instruments shall be installed within the vendor supplied venture meter or extended spool, if required without any dependency of the user.

6.5. Entire system shall be calibrated at a reputed laboratory like CEESI, SWRI.NEL etc. Calibration shall be done at 5 points and calibration report shall be provided. Customer shall have rights to witness the calibration.

6.6. Vendor shall suggest upstream and downstream requirement of Venturi based wet gas meter.

6.7. The pressure loss across a differential pressure (DP) device is a highly complex function of the properties of the fluid stream for multiphase flow. By measuring the total differential pressure (DPt) across the Venturi and the standard Venturi differential pressure (DPv), the pressure loss ratio (PLR= DPt/DRv) can be calculated. Vendor shall use the PLR together with gas to condensate ratio (CGR) to measure multi-phase flow of rates of gas, condensate and water in real time.

6.8. For BBOU, one (1) no. Pressure Transmitter, two (2) no.s of Differential Pressure Transmitter and a temperature transmitter shall be used. BBOU shall measure high Gas Volume Fraction (GVF). For ELEP, one (1) no. Pressure Transmitter, three (3) nos. of Differential Pressure Transmitter and a temperature transmitter shall be used. ELEP shall measure high Liquid Volume Fraction (LVF).

6.9. **Typical Operating Range** :Turndown >8:1, Gas mass fraction 50-100% Water liquid ratio 0-100%

6.10. **Typical Performance** : Typical Uncertainty (95% Confidence Level): • Gas mass flow rate ±2% • Condensate mass flow rate ±10% • Water volume flow rate ±1 am^3/h

6.11. **Typical Repeatability** : Gas mass flow rate <0.40%, Condensate mass flow rate <2.0%, Water sensitivity ±0.2 am^3/h

6.12. **Permanent Pressure Loss** : Pressure loss specific to application (< 1 bar)

7. Level Instruments

7.1. Level Transmitters

7.1.1. Guided Wave Radar shall be used for the measurement of level for CBD Tanks.

7.1.2. Guided Wave Radar shall be based on the Time Domain Reflectrometry (TDR) principle.

7.1.3. It shall have rigid rod of SS316 material.

7.1.4. Accuracy shall be +/- 5 mm.

7.1.5. The process connection of guided wave radar shall be 3" flanged. Guided wave radar shall be loop-powered.

7.1.6. The power supply for non-contacted type of radar transmitter, if used shall be 24VDC. The cable entries of radar shall be 1/2 NPT(F), 2 ports

7.2. Magnetic Level Gauges

7.2.1. Magnetic Level gauge shall be used for CBD vessel. Magnetic level gauge shall have IP-67 indication unit.

7.2.2. The visible portion of gauge glass shall cover the range of any associated level instrument

7.2.3. Magnetic level gauges shall be top or side mounted type and shall consist of a float with extended rod and magnet at the end of extension, magnet tube / chamber and float guide pipe. When the float moves up and down with the liquid level inside the vessel, the magnet moves up and down inside the magnet tube/chamber which in turn operates the indication mechanism.

7.2.4. The indication shall consist of bi-colour magnetic rollers mounted on outside the magnet chamber. As the float rises or falls with the liquid level each roller rotates 180 deg and so presents a contrasting colour.

7.2.5. The accidental rotation of rollers shall be prevented by the construction/design such as edge magnetisation.

7.2.6. Float failure warning shall be provided by mounting the bottom rollers of the indication with their colours reversed. In case float reaches that level, a clear readable, sharp indication will be available.

7.2.7. Vendor shall provide magnet for initial correction of bi-colour rollers.

7.2.8. Vendor to ensure that readability and dead zone errors for magnetic gauges are minimum and the same shall be highlighted in the offer.

7.2.9. Floats shall be designed and manufactured for the specific conditions specified on datasheets. It shall be designed to be adequate for hydrostatic test conditions.

7.2.10. Floats shall be hermetically sealed, no vented or pressure equalised construction shall be allowed.

7.2.11. The float, magnet tube/ chamber, extension rod, process connection flange, guide pipe shall be of SS316 unless otherwise specified.

7.2.12. The guide pipe shall be perforated, suitable for float size and shall be through out the length of float travel.

8. Fire and Gas Detection System

8.1. Gas Detection

8.1.1. Point Type and Open Path Gas flammable gas detectors and UV/IR type flame detectors shall be provided for the platforms. Point type gas detectors shall be calibrated in %LEL and open path gas detectors shall be calibrated in LEL.m.

8.1.2. Flammable gas detectors shall be set at high alarm of 25% LEL (lower explosive limit) and high high alarm of 50% LEL. For line of sight detectors settings at 1 LEL.m and 2 LEL.m are recommended.

8.1.3. A local means of visual and audible alarm shall be provided.

8.1.4. Manual Alarm Call Points shall be installed at strategic locations. Manual call points shall be explosion proof type wherever applicable according to hazardous area classification. The type of MCPs should be break glass type or pull type envisaged. In case break glass type MCPs, MCPs shall reset with test key and by replacing glass pane. Glass of MCP shall be such that it shall not injure any personnel while breaking. Spare glasses shall be provided. Once the glass of manual call point breaks, the resetting shall happen only with special keys.

8.1.5. MCPs mounted outside buildings shall be provided with FRP canopy.

8.1.6. The detector location shall be as per fire and gas system location plan.

8.1.7. All detectors shall be connected to the safety system for the reception, indication, control and relay signals originating from various fire detector or call points connected to it, for activation of fire alarm sounders and Visual alarms as per cause and effect chart.

8.1.8. The F&G system shall be designed for early and reliable detection of gas leakages, which otherwise could lead accumulations of gas, so shutdown actions can be initiated before flammable concentrations reach ignition sources. Likewise if gasses already have been ignited fires are to be detected sufficiently early so shutdown actions can be initiated to minimize risk to personnel and the plant

8.1.9. Smoke and Heat detector shall be installed inside the RTU room and shall be interfaced with SCADA system

9. Pipeline Leak Detection System

9.1. Pipeline Leak Detection

9.1.1. Fibre Optics based distributed temperature profile system shall be deployed for the leak detection as well the intrusion detection.

9.1.2. The system shall measure following parameters:

Average temperature along the sensor with spatial resolution of 1-2 m,

Average temperature threshold detection along the sensor,

Measurement of temperature variation along the sensor,

Leakage detection,

Third party intrusion detection

9.1.3. A distributed data management and analysis Software shall considered for data storing, processing, representation and analysis, as well as for the control of single or multiple reading units. The main functions of the software are aimed to measure sensors automatically. The operator shall view in real time the sensors measurement history in graphical form

9.1.4. The software shall trigger alerts (SMS, mail and phone call) and show warnings on the display. The software shall combine measurements from different sensing cables to obtain complex results.

10. Valves

10.1. ESD and BDV Valves

10.1.1. Actuated valve type, materials and connections shall be in accordance with the pipe specification. All ESD and BDV valves shall be designed as per API 6D.

10.1.2. All ESD/BDV valve shall be SIL-2/3 certified. Final SIL level shall be decided during detail engineering after SIL study.

10.1.3. The ESD / BDV valve body size shall be same as line size. Ball valve shall be used.

10.1.4. Valve design shall be Trunnion mounted ball valve, Single-body/split body with built-in double block and bleed mechanism.

10.1.5. Valve Body and Bonnet shall be meet the piping material specification and shall be generally carbon steel A105. Valve shall have 316 SS ball and trim, shall meet the piping material specification.

10.1.6. All valves shall have built-in fire-safe design.

10.1.7. All valves shall be soft seated with PTFE/PEEK material and shall be tight shut-off/ Class VI shut off.

10.1.8. Each valve shall be provided with a bolted gland type stuffing box. Packing material shall be suitable for the process temperature. In general, PTFE packing material shall be used

10.1.9. Valve connection shall be as per piping material specification. For BBOU and ELEP, all SDVs and BDVs are of 900# RTJ.

10.2. Actuators

10.2.1. Gas Over Oil (GOL) ESD/BDV valve shall be used. GOL valve shall have external gas cylinder that shall act 'instrument air' for the operation of the valve. Dry natural gas or dry air shall be used in the external cylinder. External Gas cylinder shall be equipped with Pressure Transmitter for detecting the low pressure in the cylinder and to trigger the re-fill the cylinder. External Cylinder shall be sized for the three (3) full valve cycles. Cylinder shall be designed as per ASME Section VIII and shall have pressure safety valve.

10.2.2. All ESD valve shall be 'Fail Close' and BDV valves shall be 'Fail Open'. Hand wheel shall be avoided for ESD/BDV valves.

10.2.3. Torque requirements for the actuator at any point in the stroke depend on the type of application. In selecting an actuator, to meet the torque requirements, all factors shall be taken into consideration to ensure that the delivered actuator meets the required torque output but is not significantly over-sized. Vendor shall be responsible for the right selection of the actuator based on the process data.

10.2.4. Actuator shall be sized for the shut-off differential pressure indicated in purchaser's datasheets and @ minimum pneumatic supply of gas cylinder pressure.

Service	Minimum Safety Margin (of Mas.Shut off DP) **
All ESD/BDV valves	a) SST/VSCT : 2.0 b) SRT/VRT: 1.5 c) SET/VRST:1.25 d) MAST/MST:1.1

10.2.5. Safety Margin shall be as per the Safety Margin Selection Matrix provided in the following table.

SST: Spring Start Torque

VSCT: Valve Start To Close

SRT: Spring Running Torque

VRT: Valve Running Torque

SET: Spring End Torque

VRST: Valve Re-seat Torque

MAST: Maximum Allowable Stem Torque

MST: Maximum Spring Torque.

10.3. Solenoid Valves

10.3.1. Two nos. of SIL-3 solenoid valves shall be provided and must have sufficiently sized CV so as not to impede the valve stroking time. All reliability parameters like PFD values, SFF etc shall be obtained from OEM for SIL calculation / verification during detail engineering.

10.3.2. Solenoid valve shall be explosion proof. Class H insulation allowing continuous operation in an 85°C ambient temperature shall be used in solenoid valve.

10.3.3. Quick exhaust valves shall be utilised to improve valve response time.

10.4. Valve Limit Switch

10.4.1. Limit Switches shall be Mechanical Limit Switches with SPDT signals and as per NAMUR.

10.4.2. Material of valve top box shall be Die cast aluminium with Epoxy painting.

11. Safety and Relief Systems

11.1. Safety Relief valves

11.1.1. Safety and relief valves shall meet the requirements of the ASME unfired pressure vessel code, API RP 520, and API RP 521.

11.1.2. Safety relief valve orifice sizes shall be calculated on a pure 'area' requirement basis, capacity adjustments required by code revisions or other applicable regulations shall be covered within valve selection procedure.

11.1.3. Safety valves for general process service shall be direct spring loaded with high lift characteristics and be flanged with minimum carbon-steel body and stainless steel nozzle, disc. and guides.

11.1.4. Lifting levers shall be provided as required by the applicable code for air, hot water and steam service, to allow the disc to be lifted from the seat when the operating pressure is 75% of the seat pressure.

11.1.5. Safety valves will be set to open at the pressure stated on the P&ID. Where variable back pressure exceeds 10% of the set pressure balanced bellows seal type valves shalLbe used.

12. SCADA System

12.1. Each installation (BBOU, ELEP and AKAB) shall have a SCADA System (SIL-3 PLC) that shall gather the data from the field instrument for monitoring, shutdown and communication. The SCADA shall have functional requirement of :

- PLC - Provides primary process control, monitoring, and data acquisition functions for the plant.
- SIS/ESD - Provides protection of personnel, environment, and equipment through detection and actuation of devices designed to lead the plant into a safe condition.
- F&G - Provides local and centralized warning of personnel to hand fire and gas events, through detection of gas leakage or fire.
- 2OO3 logic pressure logic shall be used for the shutdown of the flowline and opening of the corresponding blowdown valve. Voting Logic shall be implemented for Gas release and fire /flame signals.

12.2. It shall consist of plant I/O interface unit.

12.3. The scan time shall be 250 msec for close loop controls and 500 msec for open loops. For the shutdown, it shall have typically 100 ms scan time. Controller shall not be loaded more than 60% of its capacity in terms of I/O and memory, whichever is lower.

12.4. For I/O cards number of input/output channels shall be as per manufacturer standard.

It shall comprise of the following

Analogue input signals

Digital input signals

Digital output signals

Serial Interfaces

12.4.1. The analog input cards shall accept 4~20 mA DC two wire or four wire signals, 1~5VDC signals. Two wire transmitters shall be powered from the system. All digital outputs and digital inputs shall be provided with protecting fuse terminals, all analog inputs and analog outputs shall be provided with knife edge terminals in cabinet of PLC.

12.4.2. Digital input cards shall be 24VDC .All digital inputs are potential free contacts, card shall provide interrogation voltage to the input. These cards shall have channel LED for status indication.

12.4.3. Digital output cards shall be 24V DC. Interposing relay shall be used. Interposing relays shall be solid state type. Electromechnical relays are not acceptable.

12.4.4. Surge Protection Devices (SPDs) shall be used for every I/O points. Barriers shall be provided for IS signals from hazardous areas.

12.5. Engineering and Operation Station

A panel mounted engineering cum operation interface shall be provided with Emergency pushbuttons.

12.6. System Architecture and Database

12.6.1. System Architecture shall be composed of standards-based technology including latest version of approved Windows operating system with a minimum support for 10 years from the date of commissioning / hand-over to owner from Micro-soft (If Microsoft support for OS (Operating system installed for PLC is less than 10 years after hand-over date the first upgrade for OS shall be FREE OF COST to owner when support for installed OS get over (not available) from Microsoft.).

12.6.2. PLC shall comply to Open System Architecture and all communication between workstations and CPU shall be on high-speed Ethernet TCP/IP protocol according to IEEE standards.

12.6.3. The SCADA system shall maintain a database which shall allow the operator to view the information directly and indirectly related to the plant being monitored and controlled. The control system shall handle data which shall reflect:

- general alarms, group status alarms and shutdown alarms and pre-alarms.
- setpoints, measured variable and control elements status/position of all control loops
- equipment status (e.g. running, stopped, tripped, off-load, isolated for maintenance or start-up etc)
- archiving, trending and reporting, diagnostic functions.

12.7. System Integrity

The SCADA system is an essential part of the plant operability and safety. SIL-3 certified system shall be used. It shall maintain a high availability with high level of integrity. This shall be achieved by using high reliability components, hardware redundancy where appropriate and good diagnostic capability for trouble shooting.

12.8. Availability and Reliability

12.8.1. The control system shall have a high degree of tolerance to malfunction of hardware, software and operator miskeying .Any fault developed shall have its effect to localized to that part of the control system which remains operational, with a possible reduction in level of functionality. The availability of the control system for control and process monitoring shall be in excess of 99.95%.

12.8.2. Reliability data and minimum mean time between failures (MTBF) of all equipment shall be provided by OEM of PLC during detail engineering. The vendor shall provide MTBF - calculated for 50 degC cabinet internal temperature. Mean time to repair (MTTR), explains the basis of calculation, and provides reliability analysis description and relevant data. The cumulative fault repair time of the whole system shall not exceed eight hours. Moreover, the design, assembly and backup of the system shall prevent the shutdown of the whole system under any foreseeable circumstances. The Seller shall provide the actual operation availability of similar devices （A= MTBF/（MTBF ＋ MTTR)

12.9. Generation of Report

Report generation facility shall be provided for automatic and on-demand reports. All historical data shall be accessible for report generation. Automatic Reports: It shall be possible to use historical data points and generate automatically shift (8 hours), daily (24 hours), weekly and monthly reports for the purpose of plant control and plant process information. Reports shall be used for plant control contains listing of critical measurements and compressed data. It shall be possible to produce special reports for analysis purpose by operator or plant engineer using historical and compressed data poinSts on demand. All report start shall have time stamp with dates/times.

13. Programmable Logic Controllers for ESD

13.1. PLC's shall be SIL-3 Fail-safe & microprocessor based which shall be used for plant Emergency shutdown.

13.2. The system shall allow dependable and effective control of the plant and shall be designed for maximum integrity and reliability. Integrity shall be maintained by providing a fail-safe system. The system shall be TUV approved.

13.3. The system shall have facilities to replace failed equipment such as processors, I/O cards, microprocessor, and batteries on-line without a shutdown.

13.4. The system shall have high integrity including automatic crosschecking, integral self-test facility of input circuits, output circuits, and processors and provide failure indication for operator information and action.

13.5. The system shutdown shall only occur in the event of an emergency situation and not as a result of single point failure within the system. A single system fault shall not prevent correct response to a shutdown demand.

13.6. The system shall be modular in construction. The types of modules shall be kept to as minimum as possible in order to have interchangeability and low spares inventory. The system shall have extensive set of self-diagnostics for hardware and software for easy maintenance.

13.7. The system shall be immune to noise and shall ensure safe and reliable operation when subjected to RFI and electromagnetic disturbances.

13.8. The system shall be able to operate satisfactorily from 150C to 300C and 20% to 80% non-condensing humidity unless otherwise specified.

13.9. Operation of the system shall be completely unaffected if there is a momentary powerless of the order of 20 milliseconds, unless otherwise specified.

13.10. The system shall be supplied with necessary programming tools and related accessories.

13.11. The following items should be redundant:

- Controller
- Memory
- Power supply modules/units mounted in CPU and I/O rack and mounted in system cabinet
- Communication modules required for communication: Between controllers, Between controllers and HMI, between Controllers & I/O module sub system as possible
- Communication network cables and accessories.
- Servers, wherever applicable

- I/Os for critical loops / signals (to be decided during detailed engineering based on process and OEM inputs)

13.12. The PLC shall be fault-tolerant and built up in modules of standard components and software of the latest proven types and versions and shall have memory protection against power failure. Control room will be air-conditioned using split AC system. The PLC system shall have Field proven system and shall have been in use for at least 2 (Two) Years for the model offered and in similar type of application. PLC Supplier shall not offer the series of modules which will become obsolete in the next 15 years. Also, bidder shall provide support for spares & service for the plant life of 25 years from the date of handover.

13.13. Necessary REDUNDENT Cooling Fans for PLC cabinets & Air Filters with Temperature monitoring shall be provided. High Temperature & Fan Failure Alarms shall be provided by PLC vendor. The internal cabinet Temperature shall be maintained below 30 degC. Smoke and Heat detectors shall be installed inside the RTU and shall be interfaced with PLC system.

13.14. All I/O connected to control system shall have galvanic/optical isolation depending on applications. All electronic modules of PLCs shall have G3 corrosion protection as per ISA S 71.04.

14. RTU Room and Communication

14.1. The RTU room shall be located in each platform (BBOU, ELEP). For AKAB, it shall be located in the main control room.

14.2. There will be redundant Fibre Optics Links between BBOU-AKAB and ELEP-AKAB. In the failure of the fibre link, a satellite-based telecommunication system shall be automatically turned on.

14.3. The satellite-based communication shall follow ITU standard

14.4. The data from SPDC well shall be interfaced with the respective platform, namely BBOU and ELEP.

15. Utility Supplies

15.1. Instrument power supplies

15.1.1. The primary electrical supply for instrumentation shall be single-phase supply supported by an uninterruptable power supply (UPS), sized to supply emergency power on mains failure for a minimum period of 30 minutes. Power supply and UPS shall be arranged by Contractor.

15.1.2. Supplies for instrumentation shall be derived from bulk 24Vdc redundant power supply units.

16. Instrument Connections

16.1. Standard connections

16.1.1. Inline instrument process connections shall be generally flanged type to ANSI B16.5 in accordance with the piping specification.

16.1.2. Close coupled hook up shall be used.

16.1.3. Mono-flange shall be used for pressure instrument connection.

17. Instrument Cable and Wiring

17.1. Cable

17.1.1. Standard instrumentation cable shall be used in accordance with IEC standards and Shell standards.

17.1.2. All cable shall meet the requirement of IEC-60331 and IEC-60332.

17.1.3. Main cable runs between control room and field junction box shall be multipair, twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, overall screen and with 0.5mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed. Cables between Instruments and junction boxes shall be 1.5 mm2. All the multi pair/multi triad cables shall be have both individual pair screen and overall shield for both digital and analog signal cables.

17.1.4. Outer jacket colour will be:

- Black for Non-IS PLC cable
- Light Blue for IS PLC cable
- Light Blue with Gray stripes for IS ESD cable
- Red for Non-IS ESD/F&G cable
- Gray for 24VDC Non-IS Solenoid valve (ESD)
- Black for 24VDC Non-IS Solenoid valve (PLC)

17.1.5. Termination

- Terminations shall normally be of the rail mounted, preferably Wago or equivalent make.
- Both ends of all cables and their used cores shall be identified by individual cable marker rings as per drawing.
- To achieve insulation and continuity of the screen, the drain wires shall be accommodated using terminals with in the box.
- Wire colour code shall follow Shell DEP standard.
- Only continuous conductors shall be used. No splicing shall be permitted. Marking by inscription for identification on each core and sheath shall be legible and permanent.
- Normally the cores shall be 6,6 pairs/triads.
- The insulation voltage level shall be 300/500V for signal cables and 600/1000V for low voltage power cables

17.1.6. Fibre Optics Cable specification:

- A composite cable carrying 11KV power along with Fibre optics cable for communication and fibre optics for pipeline leak detection system is being planned from AKAB to BBOU and AKAB to ELEP

- Redundant Fibre Optics Links shall be established between BBOU- AKAB and ELEP-AKAB for communication.
- FOC shall be single mode 9/125 μm with wave length 1310 to 1550nm
- The design life of the optical fibre cables shall be 25 years as minimum at the specified site condition. The fibre optic cables shall have colour coding. The cable shall be jelly filled core or dry core design. The central strength member shall non-metallic and preferably FRP type. The filling compound for inside of loose tubes shall be thixotropic homogenous and shall be free from, metallic particles and non-toxic.
- Armouring shall be used for adding the mechanical strength of optical cable and this armouring shall be applied over the first jacket.
- The fibre optic cable shall have an aramid yarn layer between first jacket and wrapping layers.
- For cable protection against water ingress during storage and installation, both sides of each cable piece shall be sealed by suitable heat shrinkable end cap.
- All metal components of the joint shall be anticorrosion (stainless steel construction).
- The closure shall be re-enterable and reusable. The joint closure shall be airtight
- A complete set of required tools packed in a portable tool case for installation and re-entry shall be provided.
- Fibre joint cassettes shall be provided inside the closure to accommodate splice protector and surplus length of fibre splicing
- All Fibre optics converters / connecting components, splicing tools shall be considered.

17.2. Junction boxes (Field)

17.2.1. Junction boxes shall be certified suitable for the degree of protection IP-65, SS316, explosion proof Exd. There shall be a stainless-steel nameplate fixed permanently on the junction box.

17.2.2. Cables shall only enter from bottom. All gland plates are to be undrilled and removable. Cable glands shall be double compression type SS 316.

18. Segregation and Installation

18.1. Signal, Power and Safety cables shall run in a separate trays.

18.2. Instrument signal cables shall be run separate to power cables, the separation shall be at least 300 mm.

18.3. Intrinsically safe signal cables shall be normally routed separately to other signal types. Where this is not practical, physical separation shall be achieved by means of a metal dividing plate.

18.4. Cables shall be routed from instrument equipment rooms to field junction boxes on Stainless Steel cable tray. Cable duct and cable tray size shall be unified as following:

- 800(W)x200(H)
- 600(W)x200(H)
- 400(W)x200(H)
- 200(W)x150(H)
- 100(W)x50(H)

18.5. Control and Marshalling cabinets

18.5.1. Cabinets shall be freestanding enclosed type and shall be designed for bottom entry cables. Removable gland plates for glanding cables shall be provided.

18.5.2. Cabinets shall be fabricated from CRCA sheet with a minimum thickness of 2mm and shall be reinforced to prevent warping and buckling.

18.5.3. Cabinet doors shall be fabricated from CRCA sheet with a minimum thickness of 1.6mm. Cabinets shall be with front and rear access. Flush pull handle with keylock shall be provided to all access doors. Cabinets/Panels shall be powder coated and exterior colour shall be Siemens grey RAL 7035 and interior colour shall be glossy white. The enclosure class for cabinet outdoor shall be IP65 and the enclosure class for cabinet/panel indoor shall be IP42.

18.5.4. All cabinets shall be fitted with removable eyebolts for lifting the cabinets. Also blanking plates for eyebolts shall be provided.

19. Earthing

19.1. The following earth systems shall be provided when applicable

- Protective Earth / Electrical Earth-
- System Earth/ Signal Earth- Control Cabinet Earth (Dirty Earth) – For Signal earths, screens etc.

19.2. Protective Earth :

Earth metallic enclosure / cabinet / console etc. shall be provided with electrical earth lug, as a minimum. Unless recommended otherwise by vendor, all earthing lugs of metallic equipment shall be connected individually to electrical protective earthing system bus-bar / earthing station using a maximum of 10sq mm solid copper conductor PVC insulated wires. Where multiple cabinets are multiplexed together, earth looping with permanent shorting link cables shall be acceptable.

19.3. System Earth :

19.3.1. System earth shall be totally noise free dedicated earthing system and shall be fully isolated from electrical protective earth. This earth must be very high integrity system and shall be used to ground zero-volt references and signal cable grounds.

19.3.2. System earth shall be less than one (1) ohm grounding system recommendation with its own dedicated earth pits. The earth pits shall be suitably located outside the control room and away from any heavy noise plant equipment.

19.3.3. The earth pit design shall be as per international code of practice for earthing. A minimum of two (2) number of earth pits, in redundant configuration, shall be provided for System earth. In case number of pits required to meet system earth resistance are more than one (1) number, the number of system earth pits shall be two times the actual number of pits required to meet the redundancy requirement specified above. All these system earth pits shall be securely connected with each other to form a one homogeneous system earth grid.

19.3.4. Each marshalling / system cabinet etc. shall be provided with system earth bus-bar which shall be insulated from the metallic body frame. This bus-bar shall be used to earth also signal zero volt references and signal cable screens. Terminals used for termination of spare conductor pairs / cores of multi-pair signal / control cables shall be connected to system earth bus-bar. Shorting links shall be used for spare terminal looping.

19.3.5. System bus-bars in the multiplexed cabinets can be joined together by permanent shorting links. System bus-bars of other cabinets can also be connected together provided they are permanently joined using 35 sq. mm stranded copper conductor cable.

19.3.6. The redundant system earth pits shall be connected to the Electrical protective earthing system through suitable surge isolation and protection device like (Isolating Spark Gap- ISG) into equipotential bonding. These lightning surge isolation and protection devices shall comply to the requirements of IEC 62561-3.

20. CCTV System

20.1. IP based, PTZ night vision CCTV system shall be complete with switching unit, video server, power supply, communication equipment etc. with open connectivity, based on Ethernet/TCP-IP. The camera will be located at strategic locations in the BBOU and ELEP.

20.2. Videos from remote platforms (BBOU and ELEP) shall be sent to AKAB via redundant Fibre optics and through Satellite communication system.

20.3. The system shall offer the required functionalities from the video server and from the Company's PC through the FO communication network.

20.4. The camera along with its accessories shall be certified suitable for mounting in Classified Area as indicated in this document. The switching unit, video server, power supply & communication equipment shall be supplied by the Vendor, mounted in an operator console desk, to be located inside the RTU room which is classified as Safe Area. The operator console desk shall be supplied completely assembled with all wirings.

20.5. The complete CCTV system shall be designed such that the system shall resume its normal operation automatically, without any requirement of personnel intervention, after a power failure. The CCTV system shall provide date & time stamping on the snap shots and video recording, both in real-time and storage.

20.6. The camera and associated auxiliaries shall be mounted in Hazardous Area in an outdoor location and shall be Explosion Proof (EEx-d) certified by ATEX or CENELEC, suitable for installation in the classified hazardous area mentioned in this document.

21. Satellite Based Telecommunication System

21.1. Satellite communication shall be used for the transferring the data from unmanned platforms (BBOU and ELEP) to base station at AKAB.

21.2. Satellite communication shall be used in tandem with Fibre Optics Communication. In case of failure of Fibre Link Satellite Communication shall be automatically activated

21.3. Satellite communication shall be used for data communication, video communication, telephony system.

21.4. The system shall comprise of Antenna, Low Noise Amplifier (NA), Solid State Power Amplifier (SSPA), UP-Converter, DOWN-Converter, Routers, Modem, MUX, EPABX.

21.5. SCS shall carry the process variables like well data and flow line data. It also shall carry internet signals and CCTV video data.

21.6. The SCS system shall communicate with the SCADA systems at BBOU and ELEP and DCS system at AKAB.

21.7. SCS system shall be capable of providing the link failure alarm to SCADA and DCS system

21.8. Vendor shall suggest the uplink and downlink frequencies after studying the application and shall provide the equipment accordingly.

21.9. Cyber security aspects of the SCS shall be built-in in the SCS.

21.10. Vendor shall assist in getting all frequency license approval from the Nigerian authority.

APPENDIX - A

UNITS OF MEASUREMENT

Units shall be in accordance with the following table below :

Component	Vol%, mol%, wt ppm
Liquid flow	Kg/h,m3/h
Gas Flow	Nm3/h (0 degC, 1 atm)
Steam Flow	Kg/h
Level	%,m.mm
Pressure	Bar-g/-a,mbar-g/-a
Temperature	°C
Density	Kg/m3
Conductivity	μs/cm
Weigh	Kg, t
Viscosity	cP
Distance (Length)	m,mm
Power	W,KW
Force	N, KN
Current	A,mA
Energy	MJ,KJ
Sound	dBA
Frequency	Hz

ACCURACY TABLE FOR THE INSTRUMENT

Pressure and Differential Pressure	+/-0.075% of the calibrated span.
Temperature Transmitter	+/-0.08 degC
Wet Gas meter	• Gas mass flow rate ±2% • Condensate mass flow rate ±10% • Water volume flow rate ±1 am³/h
Guided Wave Radar	+/-5 mm

APPENDIX - B

INSTRUMENT CONNECTIONS (INDICATIVE)

Type of instrument	Equipment/Piping Connection	First Block Valve	Instrument Connection on Block valve	Instrument Connection
Part-1 Connection on instruments				
Pressure				
Pressure Gauge	1"flanged	1" flanged	**Note 1**	½" NPT(M)
Diaphragm Gauge	2" flanged	2"flanged	2" flanged	2"flanged
Pressure transmitter	1"flanged	1"flanged	1"flanged	1"flanged
Diaphragm sealed pressure transmitter	2"flanged	2"flanged	2"flanged	2"flanged
Temperature (**Note 3,6**)				
Thermowell	1 ½" flanged			1 ½" flanged
Level				
Level Indicator /Gauge	2"flanged	2"flanged	2"flanged	2"flanged
Level Switch(external float switch, side mounted)	2"flanged	2"flanged	2"flanged	2"flanged
D.P level transmitter	1"flanged	1"flanged	½"	½" NPT(F)
Diaphragm sealed level transmitter	3" Flanged	3" Flanged	3" Flanged	3" Flanged
Displacer level transmitter	3" Flanged	3" Flanged	3" Flanged	3" Flanged
Level transmitter (internal float, displacer)	4" Flanged	-	-	4" Flanged
Rader level transmitter	6" Flanged	-	-	6" Flanged

Guided wave level transmitter	3″ Flanged	-	-	3″ Flanged
Part -2 connection on the piping				
Pressure				
Pressure gauge	¾″	¾″	¾″ NPT(F)	½″ NPT(M)
Diaphragm gauge	2″ Flanged	2″ Flanged	2″ Flanged	2″ Flanged
Pressure Transmitter	¾″	¾″	¾″	½″ NPT(F)
Diaphragm sealed Pressure Transmitter	2″ Flanged	2″ Flanged	2″ Flanged	2″ Flanged
DP transmitter for flow (orifice,Venturi)	**Note 4**	**Note 4**	½″	½″ NPT(F)
Analyzer Tap(Note 5)	3″ Flanged	3″ Flanged	3″ Flanged	3″ Flanged
Temperature **(Note3,6)**				
Thermowell	1-1/2″ Flanged			1-1/2″ Flanged

Note 1 : from the block valve to instrument, the connection type meets to instruments selected.

Note 2 : the nozzle size based on the equipment or piping manufactory

Note 3 : the mentioned is only common case.

Note 4 : the nozzle on the orifice flange, the pressure of orifice flange ≤ 600#. The size of the nozzle is ½″ (DN15). The pressure of orifice flange ≥ 900#. The size of the nozzle is ¾″(DN20).

Note 5 : the detail connection will meet to the vendor of analyser

Note 6 : RTD/Thermometer ½″ NPT(M)

Above list is indicative. All pressure instruments shall have close coupled connection with double block and bleed mono-flange of the same size of first isolation valve. Details shall be indicated in hook-up diagram. Thermowell shall be 1.5″ flanged connection. Guided wave radar shall have 3″ flanged connection.

APPENDIX - C

ENVIRONMENTAL CONDITIONS

Environmental Data

Environmental Conditions

Air Temperature: 18°C to 38°C.

Relative Humidity

Parameters	Values
Mean	81%
Minimum	51%
Maximum	95%

Seismic Zone :

The project area is classified as seismic zone 0

APPENDIX – D

All instrumentation shall comply with the latest version of codes and standards of Shell DEP (latest) and international codes and standards, Shell DEP Standards :

Document Number	Document Title
DEP 30.48.00.31-Gen.	Painting and Coating of New Equipment
DEP 31.36.00.30-Gen	Pipeline Transportation Systems
DEP 31.40.00.10-Gen.	Pipeline Engineering
DEP 31.40.10.14-Gen	Pipeline Overpressure Protection
DEP 31.40.10.20-Gen	Flexible Composite Pipe System
DEP 31.40.20.37-Gen	Line pipe for critical service.
DEP 31.40.21.30-Gen.	Pipeline fittings (amendments/supplements to ISO 15590-2)
DEP 31.40.21.33-Gen.	Pig Signallers: Intrusive type
DEP 31.40.21.34-Gen.	Carbon and low alloy steel pipeline flanges for use in oil and gas operations.
DEP 31.40.60.11-Gen.	Pipeline leak detection

Discipline	Document Number	Title
Instrumentation & Control	ASME B40.1 00	Pressure Gauges and Gauge Attachments
Instrumentation & Control	ASME PTC 19.3	Thermowell Performance Test Code
Instrumentation & Control	BS EN 50288-7	Multi-Element Metallic Cables Used in Analogue and Digital Communication and Control. Sectional Specification for Instrumentation and Control Cables
Instrumentation & Control	IEC 60751	Industrial Platinum Resistance Thermometers and Platinum Temperature Sensors.
Instrumentation & Control	IEC 60079-14	Electrical Apparatus for Explosive Gas Atmospheres - Electrical Installation in Hazardous Areas (Other than mines).
Instrumentation & Control	IEC 60079-1 1	Explosive Atmospheres - Equipment Protection by Intrinsic safety.
Instrumentation &	IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)
Control		
Instrumentation & Control	IEC 61 000-6-2	Generic Standards -Immunity Standard for Industrial Environments.
Instrumentation & Control	IEC 61511	Safety Instrumented Systems for the Process Industry.

Document Number		Document Title
Instrumentation & Control	ISO 12944-2	Paints andVarnishes -CorrosionProtection of Steel Structures by Protective Paint Systems.
Instrumentation & Control	API 521	Pressure Relieving and Depressuring Systems
Instrumentation & Control	API 526	Flanged Steel Pressure Relief Valves

Additional Standards

Standards	Description
IEC 61000	Electromagnetic Compatibility (EMC)
API-RP-551	Process Measurement Instrumentation
API-RP-552	Transmission Systems
API-RP-554	Process Control System
PI-RP-555	Process Analyzer
API-RP-557	Guide to advanced control system
API 598	Valve Inspection and testing
ASME B 16.10	Face-to-face and End-to-end dimensions of valves
NAMUR	For proximity switch mounting and solenoid valve connection
ISO 5167	Measurement of fluid flow by means of pressure differential device
ASME/ANSI B16.5	Steel pipe flanges and flange fittings
ASME/ANSI B1.20.1	Pipe Threads General purpose (inch)
ISA S5.1	Instrument Symbol and identification
ISA S5.3	Distributed control and shared display unit
ISA S 5.4	Binary logic diagram for process operations
ISO 1000	International system of units
OIML R-49-1-2-3	Water Meters intended for the metering of cold portable water and hot water
OIML R-117-1	Dynamic measuring systems for liquids other than water
OIML R-137-1&2	Gas Meters
OIML R-85	Automatic level gauges for measuring the liquid in fixed storage tanks
OIML R-125	Measuring systems for mass of liquids in tanks